Learning Objective 2.6: I can measure the impedance and S-parameters of an antenna using a vector network analyzer and interpret the results.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Four names for one number.** A one-port measurement gets reported four different ways depending on which menu you open. Reference impedance is $Z_0 = 50\ \Omega$ throughout.

   (a) An antenna reads $|S_{11}| = -14$ dB at its design frequency. Find $|\Gamma|$, the VSWR, and the percentage of incident power reflected.

   $|\Gamma| =$

   VSWR $=$

   $P_{refl} =$

   (b) A second antenna shows VSWR = 2.0 at the same frequency. What is its $|S_{11}|$ in dB?

   $|S_{11}| =$

   (c) Why is -10 dB the conventional acceptance level for an antenna match? Answer in terms of power, in one or two sentences.

   (d) Lesson 7 predicted $Z_{in} = 73 + j42.5\ \Omega$ for an ideal half-wave dipole. Compute $|\Gamma|$, $|S_{11}|$ in dB, and the VSWR. Does the textbook dipole meet the -10 dB specification, and what one change fixes it?

   $|S_{11}| =$

   VSWR $=$

   Passes?

2. **From a marker readout to an impedance.** A calibrated VNA marker sitting at 1.575 GHz on a small GPS-band element reads $S_{11} = 0.33\ \angle 155°$.

   (a) Write Γ in rectangular form and find the input impedance Z_{in}.

   $Z_{in} =$ $\boxed{}$

   (b) The reactance came out positive. Is the element electrically long or short at 1.575 GHz, is its resonance above or below that frequency, and which way would you trim it?

   (c) Find $|S_{11}|$ in dB and the VSWR. Does this element meet the -10 dB specification as measured?

   $|S_{11}| =$ $\boxed{}$

   VSWR = $\boxed{}$

   (d) Suppose you trim the element perfectly, so at 1.575 GHz the reactance is zero and the resistance stays near 26 Ω . Will the match pass then? Compute it, and state what the result tells you about relying on trimming alone.

3. **Bandwidth off a sweep.** A calibrated, load-verified sweep of a wire dipole produced the following magnitudes.

f (MHz)	850	870	880	890	900	905	910	920	930	950
$\lvert S_{11}\rvert$ (dB)	-3.8	-6.5	-8.8	-12.4	-17.7	-19.4	-17.8	-12.5	-9.0	-5.2

(a) Identify the resonant frequency and say how you know.

$f_0 =$ ☐

(b) Interpolate the two -10 dB crossings and report the impedance bandwidth both in MHz and as a percentage.

BW = ☐

Fractional = ☐

(c) At resonance the Smith-chart marker sits on the real axis, to the *right* of centre. Find Z_{in} at resonance.

$Z_{in} =$ ☐

(d) A thin wire dipole typically shows 3–10% impedance bandwidth. Does this element fit the rule of thumb, and what physical change would widen it?

4. **Calibration and the reference plane.** A classmate calibrates short–open–load at the end of the test cable, then screws a 10 cm RG-58 pigtail ($v_f = 0.66$) between that connector and the antenna before sweeping.

   (a) Name the three error terms a one-port SOL calibration removes, one sentence each, and explain why exactly three standards are required.

   (b) Describe what the extra pigtail does to the measured $|S_{11}|$ magnitude and, separately, to the measured impedance. Why is this combination dangerous?

   (c) At 915 MHz, find the guided wavelength in the pigtail, its length in wavelengths, and the round-trip phase shift $2\beta\ell$ it adds.

   $\lambda_g =$ $\boxed{}$

   $2\beta\ell =$ $\boxed{}$

   (d) At what frequency is this pigtail exactly a half guided wavelength long? Why does that frequency matter to your classmate?

   $f =$ $\boxed{}$

5. **What the VNA cannot see.** The instrument reports one complex number per frequency. This problem is about the questions that number cannot answer.

   (a) You solder a 50 Ω chip resistor across the connector and sweep it. Predict $|S_{11}|$, the VSWR, and the radiated power. What does the result prove about what S_{11} measures?

   (b) Antenna A measures −25 dB with 2.0 dBi of gain; antenna B measures −9 dB with 5.0 dBi. Find the mismatch loss of each and decide which one puts more power into its main beam.

   $L_A =$

   $L_B =$

   Better:

   (c) During the lab you bring a hand within a few centimetres of the element and the dip moves. State which way the resonant frequency shifts, what happens to the depth of the dip, and why neither effect is instrument error.

   (d) Your antenna measures −30 dB at the design frequency, yet the link will not close. Name two things a VNA cannot see that would explain this, and say how you would check one of them.

Documentation: ______________________________